

Arduino-Based Object Detection, Tracking & Alert System



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Abstract

This project presents a low-cost, Arduino-based object detection and tracking system utilizing an ultrasonic sensor, servo motor, laser module, and active buzzer. The system continuously scans the environment, detects objects within a defined threshold range, and responds by pointing toward the target using the servo motor while activating visual (laser) and audio (buzzer) alerts. The project successfully simulates a simplified radar like mechanism, making it highly suitable for educational robotics, embedded systems learning, and foundational surveillance **prototyping**.

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1. Introduction

1.1 Project Overview

The integration of embedded systems in object tracking and detection plays a crucial role in modern robotics, automation, and security systems. This project mimics the fundamental principles of a radar system by employing an ultrasonic sensor mounted on a 180-degree sweeping servo motor.

1.2 Objectives

The primary objectives of this project are as follows:

- To design and implement a real-time object detection system.
- To implement servo-based directional scanning.
- To accurately detect objects using an HC-SR04 ultrasonic sensor.
- To trigger multi-modal alerts utilizing a buzzer and a laser module.
- To successfully simulate radar-like motion behavior and target locking.

2. Hardware Design & Connections

2.1 Components Used

The system relies on readily available, low-cost electronic components centered around the Arduino microcontroller.

Component	Quantity	Purpose
Arduino Uno	1	Main microcontroller processing logic.
Ultrasonic Sensor (HC-SR04)	1	Distance measurement and object detection.
Servo Motor (SG90)	1	Facilitates directional scanning.
Laser Module	1	Serves as a visual alert indicator.
Active Buzzer	1	Provides an audio alert upon detection.
Breadboard	1	Central platform for circuit assembly.
Jumper Wires	Multiple	Establishes electrical connections.
External 5V Supply	1	Ensures servo power stability (Optional but recommended).

Table 2.1: List of Components and their Purpose

2.2 Circuit Connections

Proper wiring is essential for the stable operation of both the sensors and actuators. Tables 2.2 through 2.3 detail the pin configurations.

Sensor Pin	Arduino Pin
VCC	5V
GND	GND
TRIG	Pin 10
ECHO	Pin 11

Table 2.2: Ultrasonic Sensor (HC-SR04) Connections

Module	Module Pin	Arduino / Power Pin
Servo Motor	Signal	Pin 12
	VCC	5V (External Recommended)
	GND	GND
Active Buzzer	Positive (+)	Pin 5
	Negative (-)	GND
Laser Module	Positive (+)	Pin 6
	Negative (-)	GND

Table 2.3: Actuator and Alert Module Connections

3. System Architecture & Operation

3.1 Working Principle

The core operational logic relies on a continuous mechanical sweep and acoustic ranging:

1. The servo motor sweeps continuously from 0° to 180° and back.
2. At every incremental angle, the ultrasonic sensor fires a sound pulse to measure distance.
3. If an object is detected within the defined threshold (≤ 30 cm), the servo halts at the current angle, locking onto the target. The Arduino simultaneously activates the laser and buzzer.

4. If the object moves out of range or is removed, the system deactivates the alerts and resumes scanning.

3.2 System Behavior Matrix

The state of the actuators varies strictly based on the distance feedback from the sensor, as outlined in Table 3.1.

Environmental Condition	Servo State	Laser State	Buzzer State
No object in range	Scanning (0° – 180°)	OFF	OFF
Object detected ($\leq 30\text{cm}$)	Stopped (Target Lock)	ON	ON
Object lost	Resume Scanning	OFF	OFF

Table 3.1: System Behavior Matrix

3.3 Flow Diagram

Figure 3.1 illustrates the logical execution flow programmed into the Arduino.

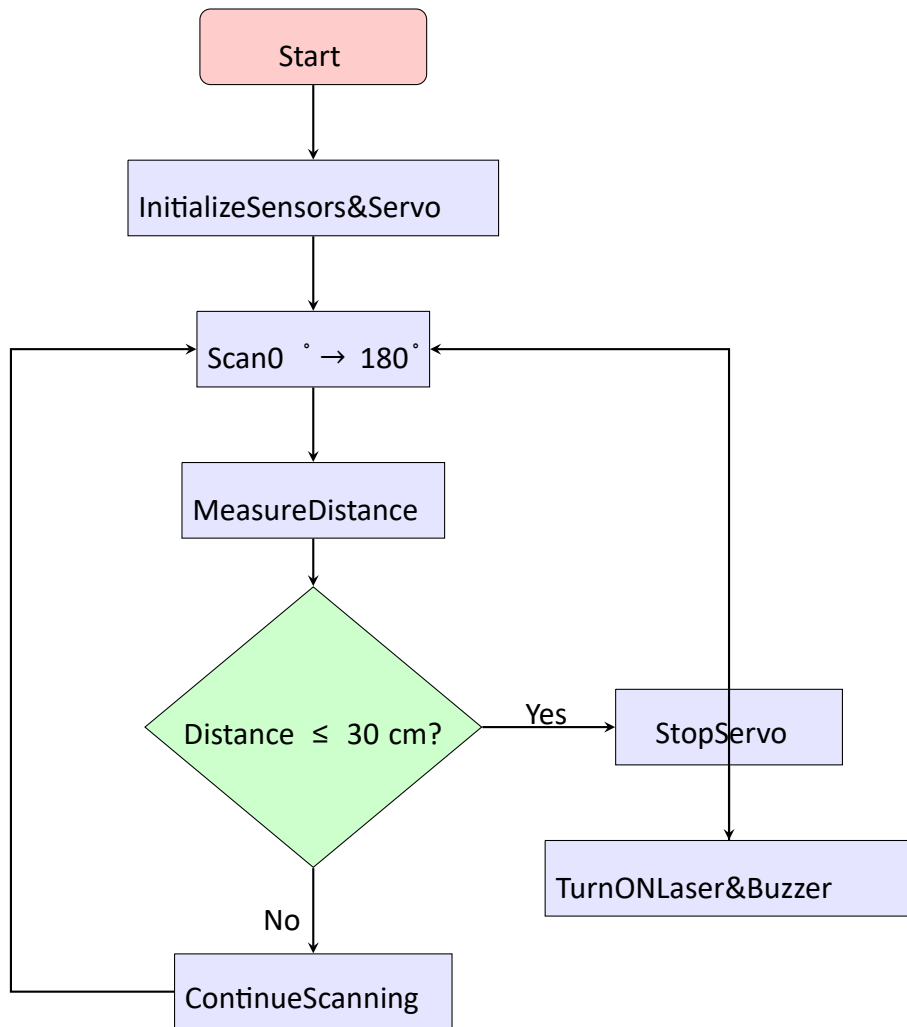


Figure3.1:LogicalFlowofOperation

4. Software Implementation

4.1 Source Code Summary

The embedded C++ code written in the Arduino IDE manages hardware interfacing. The main functional logic entails reading acoustic durations, calculating spatial distance, mapping angles to the servo, and toggling GPIO pins based on conditional statements.

4.2 Code Logic & Explanations

4.2.1 Distance Calculation Formula

To determine the distance of an object, the duration of the ultrasonic echo is multiplied by the speed of sound.

```
distance = duration * 0.034 / 2;
```

Explanation: The speed of sound in air is approximately 343 m/s, or 0.034 cm/μs. The division by 2 accounts for the sound wave traveling to the object and reflecting back.

4.2.2 Detection Condition

```
if (distance > 2 && distance < 30) {  
    // Trigger Alerts  
}
```

Explanation: This condition filters out hardware noise (distances under 2 cm, which HCSR04 struggles to measure accurately) and sets an upper limit of 30 cm for positive target identification.

4.2.3 Servo Scanning Logic

```
angle += step; servo.write(angle);
```

Explanation: Increments the servo position by a defined step, creating a smooth, radarlike mechanical sweep across the operational plane.

4.3 Project Resources & Media

All project resources including the Arduino source code, wiring schematics, and hardware images have been made publicly available on GitHub for reference and replication:

<https://github.com/AhsanSubhani3363/COAL-Final-Project->

5. Challenges & Safety Considerations

5.1 Challenges Faced

During the prototyping phase, several integration issues were encountered and resolved.

Problem Encountered	Implemented Solution
Servo motor stuttering/not moving	Fixed by implementing a dedicated external 5V power supply to prevent current draw issues on the Arduino.
Buzzer failing to emit sound	Resolved by verifying the correct GPIO pin assignment and utilizing the tone() function in code.
Unstable ultrasonic readings	Reduced the maximum detection threshold and added a lower limit to filter out electronic noise.
Erratic system behavior	Ensured a common ground connection between all modules, external power, and the Arduino.

Table 5.1: Troubleshooting and Solutions

5.2 Safety Considerations

When replicating or operating this system, the following safety protocols must be observed:

- **Laser Safety:** A low-power laser module is used. Direct eye exposure must be avoided to prevent retinal damage.
- **Electrical Grounding:** Ensure all grounds (GND) are tied together to prevent floating voltages and component damage.
- **Power Management:** Do not draw excessive current directly from the Arduino 5V pin for the servo; an external power source is highly recommended.

6. Conclusion & Future Scope

6.1 Conclusion

This project successfully demonstrates the design and execution of an automated, radarlike object detection and tracking system using an Arduino microcontroller. By integrating an HC-SR04 ultrasonic sensor with a servo motor, the prototype achieves real-time environmental scanning. The addition of a laser and active buzzer provides immediate, intuitive feedback upon target locking. The system is reliable, highly scalable, and serves as an excellent foundation for understanding closed-loop sensor-actuator relationships in robotics.

6.2 Future Improvements

While the current prototype meets all primary objectives, the system architecture can be upgraded for advanced applications:

- **Multi-Sensor Array:** Implementing multiple ultrasonic or infrared sensors for 360-degree, true-direction detection.
- **Visual Interface:** Adding an LCD or OLED display for real-time distance and angle visualization.
- **Radar UI Integration:** Interfacing the Arduino with a PC using *Processing* software to create a graphical radar screen.
- **Wireless Control:** Incorporating a Bluetooth (HC-05) or Wi-Fi (ESP8266) module for remote monitoring and manual override control.
- **AI & Vision:** Replacing the ultrasonic sensor with a camera module (e.g., ESP32CAM) running OpenCV for AI-based object classification (distinguishing between humans, animals, and vehicles).